6 SEWERAGE AND SEWAGE TREATMENT IMPLICATIONS

6.1 Introduction

6.1.1 This section presents the assessment of sewerage and sewage treatment implications impacts arising from the construction and operation of the Hung Shui Kiu New Development Area (HSK NDA).

6.1.2 The proposed preliminary alignment of the trunk sewers, including any necessary pumping provisions, for the collection, transportation, treatment and the disposal of the sewage discharged from the Developments and Infrastructure will be discussed. The sewerage and sewage treatment implications associated with the Project were assessed in accordance with Section 3.4.7 of the EIA Study Brief (No. ESB-291/2015).

6.2 Population and Employment Data

Sewerage Catchments in North-west New Territories

6.2.1 At present, the sewerage system in North-west New Territories (NWNT) serves the major urban developments in Tin Shui Wai and Tuen Mun - Yuen Long Corridor and small developments on the periphery of Yuen Long. It is delineated into two sewage catchments, namely Yuen Long Sewage Catchment and San Wai Sewage Catchment, which correspond to two of the Drainage Services Department (DSD) sewage treatment works, Yuen Long Sewage Treatment Works and San Wai Sewage Treatment Works (SW STW) respectively.

6.2.2 SW STW was planned to serve major developments in Yuen Long Town, Au Tau, Tin Shui Wai, Ping Shan, Lau Fau Shan, Shap Pat Heung and covered part of the area within HSK NDA including Hung Shui Kiu and Ha Tsuen. The remaining areas within HSK NDA are unsewered areas. Yuen Long Sewage Treatment Works serves the Eastern part of the NWNT.

6.2.3 The existing sewerage system has been reviewed under the “Adoptive Review Report” for the design and construction of Yuen Long and Kam Tin Sewerage and Sewage Disposal and the “Final Report” of the Feasibility Study for Provision of Sewerage to Unsewered Areas/ Villages in NWNT. The Upgrading of SW STW under Agreement No. CE43/2007 (DS) has been planned for completion in year 2020.

Population and employment for estimation of sewage flow

6.2.4 The project will be commissioned in different stages. A summary of estimation of sewage flow in different assessment years is summarised in **Table 6.1**.

Table 6.1 Design Population and Employment of HSK NDA for Estimation of Sewage Flow

Category	Accumulated Population / Employment		
	Year 2026(1)	Year 2031	Year 2037/38
Domestic			
- Public Rental Housing (PRH)	31,096	48,967	64,865
- Private Residential Housing	150	8,204	14,606
- Subsidised Sale Flat (SSF)	4,830	16,367	30,591
- R2 Housing	5,971	24,302	68,841
- Institutional	450	510	1,000
- Other Residents	141	141	141
Commercial			
- Hotel	637	2,170	2,170
- Sports Center	0	18	79
- Other (General – territorial average)	18,777	83,717	147,396
- Hospital	0	0	1,000
- School Student	3,225	12,497	33,707

Notes:
(1) Include the initial population intake in 2024 and population intake in Advance Works and Stage 1 of HSK NDA before 2031

6.2.5 Upon development of HSK NDA, some of the existing population and employments located within the development area under this Project will be relocated or replaced by the new development. Part of these population or employments are considered as the sewage catchment of "Upgrading of San Wai sewage treatment works – phase 1". The estimated population and employment within this sewage catchment are summarised in the **Table 6.2**. The sewage catchment that has been considered in the San Wai STW – Phase 1 and the development areas in HSK NDA are presented in **Figure 6.1**. The project details of SW STW – Phase 1 is reviewed in **Section 6.3**.

Table 6.2 Existing Population and Employment within catchment of SW STW – Phase 1 to be replaced under HSK NDA development

Category	Accumulated Population / Employment		
	Year 2026	Year 2031	Year 2037/38
Domestic			
- Traditional Village	247	755	1,032
- Other Residents	3,940	22,262	25,219
- School Students	288	2,880	3,264
Commercial			
- Manufacturing	1	4	48
- General – territorial average	627	3,517	4,277

6.3 Existing and Planned Sewerage Infrastructures

Sewerage Improvement Works

6.3.1 Recent studies on the flow projection for San Wai Sewage Catchment have been conducted under Agreement Nos. CE88/2002 (DS), CE30/2006 (DS) and CE43/2007(DS). The on-going sewerage works recommended under the above studies are tabulated in **Table 6.3**.

Table 6.3 List of On-going Sewerage Improvement Works

PWP item/Ref.	Project Description
4215DS	Sewers at Kam Tin, Tan Kwai Tsuen and Tong Yan San Tsuen – Contract No. DC/2011/07
4384DS	Yuen Long and Kam Tin Sewerage Stage 3 Package 2 – Contract No. DC/2012/05
4376DS	Trunk Sewerage at Lau Fau Shan – Contract No. DC/2011/02
4411DS	Upgrading of San Wai sewage treatment works – phase 1 – Contract No. DC/2013/10 (tentative completion in 2020)

6.3.2 However, these planned sewerage works are not designed to cater for additional flow generated from the proposed HSK NDA development.

Upgrading works of San Wai Sewage Treatment Works

6.3.3 In order to cater for the forecast sewage flow due to population growth in the NWNT, DSD have commenced the works contract no. DC/2013/10 in May 2016 - "Upgrading of San Wai sewage treatment works – phase 1" to upgrade the SW STW from preliminary treatment to chemically enhanced primary treatment plus disinfections facilities, and increase the treatment capacity to 200,000m^3 per day as shown in **Table 6.4**. The phase 1 upgrading works of the SW STW is to be carried out at a site adjacent to the existing SW STW as illustrated in **Figure 6.3**.

Table 6.4 Design Flows of SW STW – Phase 1

	Peak Wet Weather Flow (PWWF) (m^3/s)	Peaking Factor	Average Dry Weather Flow (ADWF) (m^3 per day)
Upgrading of SW STW – Phase 1(1) (tentative completion at 2020)	5.72	2.55	200,000

Notes:
(1) Refer to "DSD Agreement No. CE43/2007 (DS)"

Ha Tsuen Sewage Pumping Station

6.3.4 Under DSD Agreement CE30/2006 (DS), Ha Tsuen Sewage Pumping Station (SPS) was expanded with an additional 1200mm diameter rising main constructed in parallel with the existing twin 1200mm diameter rising mains. Upon completion of the expansion works in 2014, the capacity of Ha Tsuen SPS has been increased to 192,940m^3 per day.

NWNT Sewerage Facilities

6.3.5 The effluent of the upgraded SW STW will be discharged to the existing junction chamber of the 9km long NWNT tunnel (of capacity 11.6m^3/s in terms of PWWF) and eventually discharged to the existing Urmston Road Outfall. The alignment of the existing NWNT effluent tunnel is shown in **Figure 6.2**.

6.3.6 With reference to the Operation and Maintenance Manual for NWNT Sewerage Scheme, the existing Urmston Road Outfall has been designed for a PWWF of 5.1m^3/s with a peaking factor of 1.5 taking into consideration that the large volumetric capacity of the 9km long NWNT effluent tunnel will attenuate the flow feeding into the downstream system. **Table 6.5** illustrates the design flow and capacity of the above NWNT Sewerage Facilities.

Table 6.5 Design Flows and Capacity of NWNT Sewerage Facilities

NWNT Sewerage Facilities	PWWF (m^3/s)	Peaking Factor	ADWF (m^3 per day)
NWNT Effluent Tunnel	11.6	2.5	400,896
Urmston Road Submarine Outfall	5.1	1.5	294,624

Notes: refers to "Operation and Maintenance Manual of North West New Territories Sewerage Scheme"

6.4 Design Criteria

Unit Flow Factors

6.4.1 The average dry weather sewage flows of the HSK NDA sewage sub-catchments are calculated based on the latest development schedule of the Revised RODP of HSK NDA with unit flow factors given in the EPD Report No. EPD/TP 1/05 Guidelines for Estimating Sewage Flows (GEFS) Ver 1.0.

6.4.2 The unit flow factors (UFFs) and population together with appropriate peaking factors have been adopted in the flow projection and hydraulic assessment and discussed in this section.

6.4.3 The UFFs adopted in the assessment for HSK NDA are shown in **Table 6.6**:

Table 6.6 Unit Flow Factors Adopted for HSK NDA

Category	Unit	Unit Flow Factors (m^3 per day)
Domestic		
• Traditional Village	person	0.15
• Public Rental House (PRH)	person	0.19
• Private Permanent Housing	person	0.37(6)
• Subsidised Sales Flat (SSF)	person	0.27(7)
• R2 Housing	person	0.27
• Institutional and Special Class	person	0.19
• Other Residents (General- Permanent housing)	person	0.25(1)
Commercial and Students		
• Hotel	employee	1.58(5)
• School student	person	0.04
• Hospital	employee	0.37(3)
• Sports Complex	employee	0.35(3)
• Others (General – territorial average)	employee	0.28(4)(5)
Industrial		
• Manufacturing	employee	2.784(2)

Notes:
(1) UFF of domestic housing for Yuen Long is referenced from EPD GESF Table T-1.
(2) UFF is referenced from CE 43/2007 (DS) "Working Paper on Design Flow, Loads, Capacity, Treatment Level and Performance Standards" and for the design of Upgrading of San Wai STW.
(3) UFF is referenced from CE 35/2006 (CE) "Kai Tak Development – Final Sewerage Impact Assessment Report".
(4) UFF of commercial activities "General – territorial average" is referenced from EPD GESF Table T-2.
(5) UFF including 0.08 for "commercial employee" is referenced from EPD GESF Table T-2.
(6) For "Private Permanent Housing" with residential development at HSK NDA, Type R3 or R4 unit flow factor is adopted.
(7) For "SSF" with residential development at HSK NDA, Type R2 unit flow factor is adopted.

Peaking Factors

6.4.4 The peak flow for the corresponding section of trunk sewer is used in the hydraulic assessment to examine the sewerage impact. The peak flow is calculated from multiplying the ADWF by the peaking factor.

6.4.5 The peaking factor is based on the cumulative contributing population. The contributing population is equal to the calculated total average flow divided by a factor of 0.27. The factor is taken as an average unit flow factor described in Paragraph 12 of the EPD GEFS Ver 1.0 from EPD. As the cumulative contributing population increases, the peaking factor and the peak flow will decrease.

6.4.6 Although separate sewerage system and stormwater system will be designed for HSK NDA, as a conservative approach, the peaking factors including stormwater allowance was used for assessing the peak flow condition. The peaking factors were selected in accordance to EPD GEFS Ver 1.0 as shown in **Table 6.7** below:

Table 6.7 Peaking Factor for Sewers in HSK NDA

Population Range	Global Peaking Factor (including stormwater allowance) for sewers
< 1,000	8
1,000 – 5,000	6
5,000 – 10,000	5
10,000 – 50,000	4
> 50,000	Max [$7.3/N^{0.15}$, 2.4]
Population Range	**Global Peaking Factor (including stormwater allowance) for STW and SPS**
< 10,000	4
10,000 – 25,000	3.5
25,000 – 50,000	3
> 50,000	Max [$3.9/N^{0.065}$, 2.4]

Note: N is the contributing population in thousands

Pipe Roughness

6.4.7 The pipe roughness is specified as an equivalent sand roughness (ks) used by the Colebrook-White equation. In this hydraulic assessment, the roughness coefficient used for the existing sewer would follow the values (ks=3mm) adopted in the previous Review of Yuen Long and Kam Tin Sewerage and Sewage Treatment Requirements.

6.4.8 The roughness coefficients used for the proposed sewer are listed in **Table 6.8** below:

Table 6.8 Pipe Roughness Coefficient used in HSK NDA

Pipe Size	Pipe Material	ks
<600mm Gravity Sewer	Vitrified Clay Pipe	0.6mm
>=600mm to 2400mm Gravity Sewer	Precast Concrete Pipes with PVC lining	1.5mm
Rising Main	Mild Steel Pipe	0.15mm

Sediments

6.4.9 The presence of sediments inside the trunk sewer will reduce the capacity of the sewer due to the reduction in cross sectional area. In this hydraulic assessment, the sediment depths to be used for the existing sewer will follow that adopted in the previous Review of Yuen Long and Kam Tin Sewerage and Sewage Treatment Requirements which is listed below:-

- Siltation = 5% for gradient larger than 1 in 25
- Siltation = 10% for other cases

Hydraulic Assessment Methodology

6.4.10 The hydraulic assessment methodology is composed of two portions:-

- Hydraulic Assessment Methodology for the New Sewers inside HSK NDA; and
- Hydraulic Assessment Methodology for the Existing Trunk Sewer.

Modelling Approach

6.4.11 With the above methodologies, the following approach is applied for the hydraulic modelling assessment without availability of calibrated flows data for the existing trunk sewer:

i. Identify the discharge points from the existing sewage catchments and the HSK NDA sewage catchments for assessment.

ii. Calculate the ADWF of the sewage catchments.

iii. Determine the peaking factors based on the cumulative contributing population at different sections of the trunk sewer in different scenarios.

iv. Calculate the peak flows for different sections of the trunk sewer in different design scenarios.

v. Enter inflow hydrographs with the peak flows to the model in different design scenarios at different sections of the trunk sewer for model runs.

vi. Conduct model runs in different design scenarios.

Model Networks and Catchment

6.4.12 The following different peaking factor scenarios were applied for hydraulic analysis:-

- Different Global Factor Scenarios for different sections of sewers; and
- Performance Scenario (a flat DWF profile of 3 times ADWF for simulation runs with duration not less than 3 hours; and a flat DWF profile of 4 times ADWF with duration of 2 hours for sewer size smaller or equal to 375mm diameter).

6.4.13 The Performance Scenario used for hydraulic analysis was made reference to Chapter 11 of the EPD Guidelines for Sewer Network Hydraulic Model Build and Verification for the new sewers inside HSK NDA.

6.4.14 The modelling approach was based on the EPD Guidelines for Sewer Network Hydraulic Model Build and Verification for the hydraulic modelling assessment for the new sewers inside HSK NDA.

Model Simulation

6.4.15 InfoWorks was used as a tool in the hydraulic assessment. InfoWorks has the capability of modelling urban sewerage systems and is capable of determining the capacity of culverts, pipes, channels and hydraulics structures. Results from the sewerage model are considered more representative to reflect a realistic scenario than conventional desktop empirical calculations.

6.5 Estimation of Sewage Flow

Net additional sewage flow due to new development of HSK NDA

6.5.1 Based on the population and employment data in the HSK NDA development, the sewage flow generated from design population of HSK NDA will be 91,245m^3 per day. With deduction of the sewage from replacement of existing population and employment in the development areas, which have been considered in San Wai sewage catchment, the net additional sewage flow due to new development of HSK NDA project will be **85,408 m^3 per day** and will be handled in the new HSK STW in HSK NDA. The detailed sewage figures are presented in the **Table 6.9.**

Table 6.9 Estimated Net additional in Sewage Flow arising from HSK NDA development

Category	Accumulated Projected ADWF (m^3 per day)		
	Year 2026(2)	Year 2031	Year 2037/38
Sewage Flow from design population of HSK NDA			
Domestic			
- Private Resident	1,668	9,597	23,991
- Public Housing (PRH and SSF)	7,212	13,723	20,584
- Institutional	86	97	190
- Other Residents (General- Permanent housing)	35	35	35
Commercial			
- Others (General – territorial average)	5,257	23,441	41,271
- School Student	129	500	1,348
- Hotel	1,007	3,428	3,428
- Sports Center	0	6	28
- Hospital	0	0	370
Total Sewage Flow from HSK NDA…(a)	15,394	50,827	91,245
Category	**Accumulated Projected ADWF (m^3 per day)**		
	Year 2026(2)	**Year 2031**	**Year 2037/38**
Sewage Flow originally handled by San Wai STW – Phase 1 within HSK NDA (to be replaced upon HSK NDA Development)			
Domestic			
- Traditional Village	37	113	155
- Other Residents (General- Permanent housing)	985	5,565	6,305
Commercial			
- Others (General – territorial average)	176	985	1,198
- School Student	12	115	131
Industrial			
- Manufacturing	2	12	133
Total Sewage Flow …(b)	1,211	6,791	7,920
Net additional sewage flow due to the new development of HSK NDA			
Total Sewage Flow(1) …(a)-(b)	14,538	45,137	85,408

Notes:
(1) Include a minor portion of 2.5% return liquid allowance.
(2) Include the initial population intake in 2024 and population intake in Advance Works and Stage 1 of HSK NDA before 2031

6.6 Implementation of HSK STW

Proposed HSK STW for HSK NDA

6.6.1 As the planned SW STW – Phase 1 was not designed to cater for the additional flow generated from the HSK NDA development, a new HSK STW is proposed to handle the sewage flow of 85,408 m^3 per day. For the design purpose, the HSK STW (Phase 1) is proposed to provide treatment capacity of 70,000 m^3 per day using the Site 3-26 in the Revised RODP and the HSK STW will be expanded to HSK STW (Phase 2) with total treatment capacity of 85,500 m^3 per day. As the SW STW – phase 1 is designed to replace the role of the existing SW STW to cater for the sewage in San Wai catchment, part of the land of existing SW STW will be explored as a potential location for the expansion of HSK STW (Phase 2) subject to future review of population growth in the San Wai Sewage Catchment and the TPEDM data at that time. A separate EIA Study under Schedule 2 of EIAO will be conducted for the new HSK STW.

6.6.2 The HSK STW (Phase 1) with treatment capacity of 70,000 m^3 per day is proposed to commission in 2031 to serve the HSK NDA. A further study and review on the updated population data from later version of TPEDM and the sewage flow on San Wai Sewage Catchment is recommended to ascertain the required capacity for expansion to HSK STW (Phase 2) to serve the ultimate development of HSK NDA. .

6.6.3 In the early stage of the HSK NDA development, the initial sewage flow from the Advance Works and Stage 1 is proposed to be temporarily treated by the SW STW – Phase 1 that, upon advice by EPD, will have adequate capacity before 2031. The proposed development sites of Advance Works (ASEP) and Stage 1 are presented in the **Figure 2.19**. The estimated sewage flow is around 14,538 m^3 per day and tabulated in year 2026 of **Table 6.9**. Upon commissioning of proposed HSK STW (Phase 1) in 2031, all sewage arising from HSK NDA including the above-mentioned initial sewage flow will be solely handled by the new HSK STW (Phase 1). The sewage flow of about 7,920 m^3 per day originally handed by the upgraded SW STW – Phase 1 within HSK NDA will be handled by the upgraded SW STW – Phase 1 ultimately.

6.6.4 The option of on-site treatment to handle sewage from initial population intake of HSK NDA in 2024 before commissioning of HSK STW in 2031 has also been considered. As compared with the option of using SW STW - Phase 1, this option will reserve more land within individual sites for temporary sewage treatment facilities and will induce nuisance and water quality concern during operation of treatment facilities near the residential areas. In addition, the deployment of temporary treatment facilities is not a cost-effective solution. This interim measure would also be abandoned in later development stages and the disposal of these temporary on-site treatment facilitates may induce air, water quality and waste issues. In view of the above operational and environmental concerns and available spare capacity of SW STW - Phase 1 before 2031, the option of using SW STW (Phase 1) can be considered as a temporary measure.

NWNT Sewerage Infrastructure

6.6.5 The effluent of the upgraded SW STW and the HSK STW will be discharged to the existing NWNT Effluent Tunnel which have total capacity 400,896 m^3 per day as shown in **Table 6.5**. This tunnel is connected to the existing submarine outfall at Urmston Road.

Effluent Discharge and Reuse of Reclaimed Water in HSK NDA

6.6.6 In order to achieve a sustainable development, reclaimed water treated from the proposed HSK STW will be reused for toilet flushing in HSK NDA.

6.6.7 The demand of flushing water under HSK NDA is estimated as 35,824 m^3 per day and will be provided by tertiary treatment from the proposed HSK STW to ensure higher water quality standard while the rest of sewage effluent is suggested to be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to Urmston Road Outfall. The need for a higher treatment level than secondary plus treatment will be considered with further review on water quality condition in future separate EIA for the HSK STW.

6.6.8 Estimated volume and target water quality standard of reclaimed water which will be subject to further study and design of new HSK STW, are shown in **Table 6.10** and **Table 6.11**. The target water quality standard is for reference and subject to further review in future review and EIA Study under Schedule 2 of EIAO for the HSK STW.

Table 6.10 Estimated Volume of Reclaimed Water Reuse and Disposal of Treated Sewage

Description	Volume (m^3 per day)
Total sewage flow	85,500
Reclaimed Water reused in HSK NDA for flushing (Tertiary Treatment)	39,804 (35,824/0.9*)
Remaining Treated Sewage to be discharged to Urmston Road Outfall (Secondary Plus Treatment)	45,696

Note:
* With reference to sewage treatment projects regarding reclaimed water generation including NENT project, minimum 90% of treated sewage can be reused.

Table 6.11 Target Water Quality Standard for Reclaimed Water for Flushing

Parameter	Unit	Reclaimed Water Quality(1)
E. Coli	cfu/100mL	Not detectable
Total residual chlorine (TRC)	mg/L	≥ 1 (out of treatment system) ≥ 0.2 (at point-of-use)
Dissolved oxygen (DO)	mg/L	≥ 2
Total suspended solid (TSS)	mg/L	≤ 5
Colour	unit	≤ 20
Turbidity	NTU	≤ 5
pH	-	6 – 9
Threshold odour number (TON)	-	≤ 100
5-day Biochemical oxygen demand (BOD5)	mg/L	≤ 10
Ammoniacal nitrogen (NH3-N)	mg/L as N	≤ 1
Synthetic detergents	mg/L	≤ 5

Notes:
(1) The parameters are referenced to the target water quality standard adopted in North East New Territories New Development Areas (NENT NDA) project.

6.6.9 With reference to the Environmental Impact Assessment report approved in 2003 for Upgrading and Expansion of SW STW, the upgraded SW STW (Phase 1) equipped with Chemical-enhanced Preliminary Treatment (CEPT) with disinfection will improve the effluent quality. The water quality impact due to proposed effluent discharge of 246,000 m^3 per day of CEPT treated effluent had been assessed and the impacts are within an acceptable level. Upon the reuse of reclaimed water in HSK NDA, the quantities of effluent discharge in this Project via Urmston Road Outfall are summarised at **Table 6.12**.

6.6.10 As smaller quantities of effluent will be discharged via Urmston Road Outfall and secondary plus treatment level instead of CEPT with disinfection will be provided in the new HSK STW, it is envisaged that the water quality would be further improved as compared with the condition in the approved EIA.

Table 6.12 Summary of Effluent Discharge from HSK NDA and San Wai STW – Phase 1

Total Treated Sewage Effluent from HSK NDA to be discharged (with Secondary Plus(1) Treatment)	SW STW – Phase 1 with CEPT (m^3 per day)	Discharge to Urmston Road Outfall (m^3 per day)
45,696	200,000	200,000 (CEPT) + 45,696 (Secondary Plus)

Note:
(1) Secondary Plus Treatment is a combination of secondary treatment with 75% nitrogen removal and UV disinfection.

6.6.11 The land for the HSK STW (Phase 1) and service reservoir(s) for reclaimed water have been reserved in the Revised RODP. The land requirement for the new HSK STW (Phase 1) and further expansion to Phase 2 are approximately 4.0 ha and 1.0 ha respectively.

6.7 Sewerage Impact

Ha Tsuen Sewage Pumping Station (Ha Tsuen SPS) and New Sewage Pumping Stations

6.7.1 It is noted that the expanded Ha Tsuen SPS (with capacity of 192,940 m^3 per day) is to serve the upgraded SW STW (Phase 1) with an estimated sewage flow of 200,000 m^3 per day as listed in **Table 6.4**. Thus, provision of new SPS(s) servicing the new HSK STW is required to cater for the additional sewage flow from HSK NDA.

6.7.2 In accordance with new developments listed out in the Revised RODP, the existing Ha Tsuen SPS and new pumping stations, namely SPS1, SPS2, SPS3 and SPS4, are proposed to convey the sewage flows to the upgraded SW STW (Phase 1) for existing flow or HSK STW for new flow. The locations and catchments of the existing and proposed sewage pumping stations are shown in **Figure 6.4**. SPS2 will receive the flow pumped from its own catchment area from SPS1 and SPS3 and then convey to HSK STW. However, the existing flow originally handled by SW STW (Phase 1) will still be treated by the SW STW. The projected sewage flow for each SPS is summarised in **Table 6.14**.

6.7.3 The existing Ha Tsuen SPS is proposed to collect the sewage in the eastern part of HSK NDA. Together with the original sewage flow to be handled by Ha Tsuen SPS, the sewage flow will exceed the capacity of Ha Tsuen SPS at Year 2037/38 (with capacity of 192,940 m^3 per day) as presented in Table 6.13 below. To spare the capacity of Ha Tsuen SPS, it is proposed to divert the flow of existing sewage catchment from Tin Wah Road SPS to SPS4 and eventually to new HSK STW and SW STW (Phase 1) respectively in 2031 as shown in **Figure 6.3**. On the other hands, the SPS4 will convey the equal amount of sewage flow of 30,332 m^3 per day from Tin Wah Road SPS to the new HSK STW so as to compensate the sewage flow that will be collected in the Ha Tsuen SPS for HSK NDA. Therefore, the SW STW (Phase 1) will not handle any additional sewage flow from the HSK NDA. In this connection, the sewerage works from Ha Tsuen SPS to the HSK STW to convey the sewage from Ha Tsuen SPS can be avoided for cost effectiveness.

6.7.4 This diversion of rising mains to SPS4 will not only spare the capacity in Ha Tsuen SPS for the sewage flow from HSK NDA but also avoid the maintenance and management problem on the twin 800mm rising mains as Tin Ying Road between its junction with Tin Wah Road and Ping Ha Road will be removed and replaced by development sites. As advised by Drainage Services Department, the design capacity of Tin Wah Road SPS is 43,546 m^3 per day. The capacity checking on the Ha Tsuen SPS is shown in **Table 6.13**.

Table 6.13 Capacity Checking on Ha Tsuen SPS

Catchment Area	Accumulate Sewage Flow (m^3 per per day)(1)		
	Year 2026	Year 2031	Year 2037/38
Ha Tsuen SPS (Total) (2) (**without** diversion of sewage of Tin Wah Road SPS)	180,430	181,843	202,133
Ha Tsuen SPS (Total) (2) (**with** diversion of sewage of Tin Wah Road SPS to SPS 4 in 2031)		138,297	158,587

Notes:
(1) Include a minor portion of inflow from dry weather flow interceptors (i.e.: 10,363 m^3 per day) and 2.5% return liquid allowance.
(2) Include sewage flow due to HSK NDA.

6.7.5 **Table 6.14** to **Table 6.16** show the accumulated projected sewage flow, peaking factors and accumulated projected peak flows respectively for SPS1, SPS2, SPS3 and SPS4 for Year 2026, year 2031 and Year 2037/38.

Table 6.14 Estimated Sewage Flows for Year 2026, Year 2031 and Year 2037/38 Scenarios

Facilities	Accumulated Sewage Flow (m^3 per day) (1)		
	Year 2026	Year 2031(4)	Year 2037/38(4)
Pumping Station SPS1	4,067	26,641	26,641
Pumping Station SPS2(3)	5,051	36,502	39,130
Pumping Station SPS3	NA	8,329	10,957
Pumping Station SPS4	NA	49,430 (2)	67,943 (2)

Notes:
(1) Include 2.5% return liquid allowance.
(2) Include sewage from Tin Wah Road SPS.
(3) SPS 2 will handle the sewage from SPS 1 and SPS 3
(4) From 2031 and onward years, the sewage flow from HSK NDA will be conveyed to the new HSK STW.

Table 6.15 Peaking Factors for Pumping Stations in HSK NDA

Facilities	Peaking Factors (PF) (1)		
	Year 2026	Year 2031	Year 2037/38
Pumping Station SPS1	3.27	2.90	2.90
Pumping Station SPS2	3.23	2.84	2.83
Pumping Station SPS3	NA	3.13	3.07
Pumping Station SPS4	NA	2.78	2.73

Notes:
(1) Peaking factors (PF) are calculated from EPD's guideline for population>50,000:
Max ($3.9/N^{0.065}$, 2.4), where N is the contributing population in thousands
Contributing population = Calculated total average flow in m3 per day /0.27/1000.

Table 6.16 Estimated Peak Flows for Year 2026, Year 2031 and Year 2037/38 Scenarios

Facilities	Accumulated Projected Peak Flows (m^3/s) (1)		
	Year 2026	Year 2031	Year 2037/38
Pumping Station SPS1	0.15	0.89	0.89
Pumping Station SPS2	0.19	1.20	1.28
Pumping Station SPS3	NA	0.30	0.39
Pumping Station SPS4	NA	1.59	2.14

Notes:
(1) Projected Peak Flows = (Projected ADWF x PF) x (1+2.5%).

Hydraulic Assessment

6.7.6 Hydraulic model was established to assess the hydraulic of the recommended sewerage systems developed according to the assessment methodology in **Section 6.4**.

6.8 Implementation of Proposed Sewerage Works in HSK NDA

Introduction

6.8.1 The proposed sewerage works within HSK NDA have been highlighted in **Section 6.7** and are recommended to divide into **two** principal phases below based on the development phasing of HSK NDA.

6.8.2 **Figure 6.2** presents the existing sewerage network inside HSK area.

Phase A Sewerage Works

6.8.3 Phase A Sewerage Works comprises construction of two new sewage pumping stations, SPS1 and SPS2, and its associated pipe works is the major activity of Phase A Sewerage Works in order to cope with the Advance Works and Stage 1 of HSK NDA.

6.8.4 SPS1 is designated to convey sewage from southern area of HSK NDA next to Castle Peak Road to SPS 2 and then to SW STW (Phase 1) temporarily until the commissioning of HSK NDA in 2031. SPS2 sewage catchment also includes western area of HSK NDA next to Kong Sham Western Highway.

6.8.5 The proposed sewerage systems are designed in accordance with the methodology stated in **Section 6.4** for the new sewers inside HSK NDA. The proposed sewerage networks for the Phase A Sewerage Work are recommended to be completed by 2026 to cope with the Advance Works and Stage 1 development, see **Figure 6.5**.

Phase B Sewerage Works

6.8.6 In phase B sewerage works, the new HSK STW will start commissioning to cater the sewage flow from HSK NDA from the four proposed sewage pumping stations. In this connection, the rising mains from SPS 2 will be disconnected from SW STW (Phase 1) and connected to the new HSK STW as shown in **Figure 6.6**.

6.8.7 The sewerage works in this phase comprises construction of new sewage pumping stations, SPS3 and SPS4 and diversion of the existing twin 800mm diameter rising mains along Tin Ying Road. SPS3 will receive sewage flow from middle area of HSK NDA and then convey them to SPS2 for further pumping to the new HSK STW.

6.8.8 New rising mains and gravity sewers will be constructed to connect SPS3 to SPS2 for Stage 2 and later development stages of HSK NDA.

6.8.9 Twin rising mains with 800mm diameter will be constructed in order to convey sewage from Tin Wah Road SPS to SPS4 and eventually to SW STW (Phase 1) while the existing twin 800mm diameter rising mains from Tin Wah Road SPS to Ha Tsuen SPS along Tin Ying Road will be abandoned to correspond with the proposed removal of Tin Ying Road.

6.8.10 In addition to conveying sewage from Tin Wah Road SPS to SW STW (Phase 1) via SPS4, SPS4 will collect sewage from northern area of HSK NDA and pump to the new HSK STW through a proposed new separate rising main from SPS4 to HSK STW.

6.8.11 The proposed sewerage networks for the Phase B Sewerage Works are recommended to be completed by 2031. The key sewerage works in Phase B are presented in **Figure 6.6**.

6.9 Cumulative Impact from Concurrent Project

6.9.1 As a separate sewerage system and a new STW are proposed to cater net additional sewage flow arising from HSK NDA and no sewerage system from other concurrent projects will affect or be affected by this Project. Therefore, no adverse cumulative sewerage and sewage implication would be expected.

6.10 Environmental Acceptability of Schedule 2 Designated Projects

6.10.1 An application for EP would be submitted under this EIA for DP1, DP2, DP5, DP6, DP9 and DP12 as presented in **Figure 1.3**.

Sewage Pumping Station (DP9)

6.10.2 With proper implementation of the proposed sewerage system and proposed pumping stations (as detailed in **Sections 6.7 and 6.8** above) which will be implemented in two phases, no sewerage impact would be resulted from the proposed NDA development.

New Roads (DP1, DP2, DP5, DP6 and DP12)

6.10.3 The water quality impacts arising from sewage generated for construction activities have been assessed in the Chapter 5 – Water Quality Impact. No sewerage and sewage impact is anticipated in the operation phases of the proposed road in the DPs of new roads.

Other DPs

6.10.4 There will be separate EIA studies to assess the following Schedule 2 DPs (refer to **Figure 1.2**):

- DP3 - Construction of new West Rail HSK Station (Site 4-34)
- DP4 - Construction of EFTS – subject to further review
- DP7 - Construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) – subject to further review
- DP8 – Construction of new HSK STW (Site 3-26 and part of existing SW STW)

- DP10 - Construction of Flushing Water Service Reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40)

- DP11 - Construction of one RTS (Site 3-12)

6.10.5 The detailed sewerage and sewage impact of these Schedule 2 DPs including the water quality impacts from the new HSK STW will be further investigated in separate EIA studies under the EIAO. The EM&A requirements for these Schedule 2 DPs will also be provided under separate EIA studies.

6.11 Summary

6.11.1 With sewage estimation based on population data in the Revised RODP and population to be replaced in the HSK NDA being covered by SW STW – Phase 1, the net additional sewage flow from HSK NDA is **85,408 m^3 per day** in ultimate development.

6.11.2 A new HSK STW (Phase 1) will be provided with treatment capacity of **70,000 m^3 per day** using the land reserved in the HSK NDA. Further expansion to new HSK STW (Phase 2) will provide a total treatment capacity of **85,500 m^3 per day** which may consider utilising part of the land of the existing SW STW, after commissioning of San Wai STW – Phase 1. The sewage will be treated by tertiary treatment and secondary plus treatment (with UV disinfection and 75% nitrogen removal). Further study and review on the updated population data and the sewage flow on San Wai sewage catchment will be carried out to ascertain the required capacity for the HSK STW (Phase 2). A separate EIA Study for the new HSK STW will be conducted under Schedule 2 of EIAO.

6.11.3 The associated sewerage facilities such as rising mains and pumping stations are proposed. The proposed sewerage facilities layout plan is shown in **Figure 6.3** along with existing and planned sewerage system.

6.11.4 Reuse of reclaimed water is recommended for non-potable uses such as toilet flushing. The reclaimed water will be provided from the HSK STW with tertiary treatment to ensure a higher water quality standard. The rest of sewage effluent will be treated by secondary plus treatment and discharged to Urmston Road Outfall.

6.11.5 Based upon a preliminary sewerage impact assessment as described in this section, it can be concluded that the proposed development is sustainable from sewerage collection, treatment and disposal perspective.